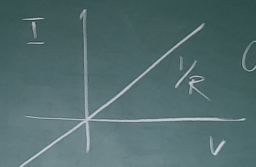
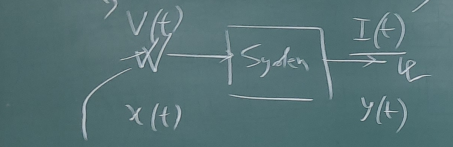
$q = CV$ $\quad I = C\frac{dV}{dt}$ (PS1)

R



Ohm's law



$x_1 \to y_1$

$x_2 \to y_2$

Linear system

$\alpha_1 x_1 + \alpha_2 x_2$

$\to \alpha_1 y_1 + \alpha_2 y_2$

for $\forall \alpha_1, \alpha_2$

(PS2)

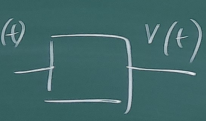
I/P V(t)

O/P I(t)

Capacitor (system) is linear

I/P $\quad I(t)$

O/P $\quad V(t)$



$$I(t) = C\frac{dV(t)}{dt}$$

$$\Rightarrow V(t) = \int_0^t \frac{1}{C} I(u)\,du + V(0)$$

$\alpha_1 V_1(t) + \alpha_2 V_2(t)$

$$= \alpha_1\left(\frac{1}{C}\int_0^t I_1(u)\,du + V_1(0)\right) + \alpha_2\left(\frac{1}{C}\int_0^t I_2(u)\,du + V_2(0)\right)$$

$I_1(t) \to V_1(t)$

$I_2(t) \to V_2(t)$

$\alpha_1 I_1(t) + \alpha_2 I_2(t)$

?? $\longrightarrow \alpha_1 V_1(t) + \alpha_2 V_2(t)$

Linear if the initial voltage = 0 !

Pg 4

Capacitor is NOT exactly linear, but almost linear

Laplace-domain (resistor + battery)
↓
source

$x(t)$ → [$h(t)$] → $y(t)$

$Y(s) \leftrightarrow y(t)$

$H(s) \leftrightarrow h(t)$

LTI

(One-sided) laplace transform

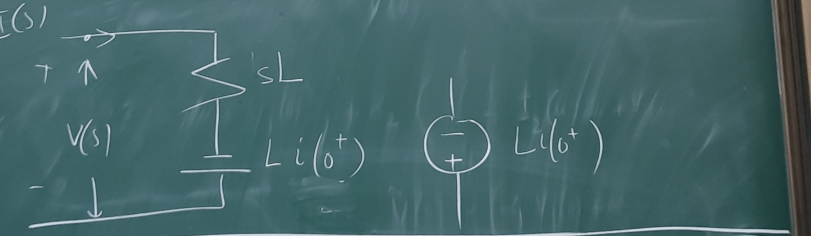
$$y(t) = \int x(u)\, h(t-u)\, du$$

$$X(s) = \int_0^\infty e^{-st}\, x(t)\, dt$$

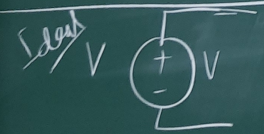
s complex

Pg 9

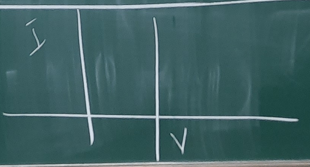
$$V(s) = L\left(sI(s) - i(0^+)\right)$$



Ideal



can draw any current

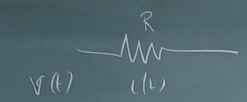


PDF

$Y(s) = X(s)\,H(s)$ for LTI

$H(s) = \frac{1}{s+a}$

$H(s) = \frac{1}{s}$



$v(t) = R\,i(t)$

$\mathcal{L}$

$V(s) = R\,I(s)$

$H(s)$ is a constant R

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$i(t) = C\frac{dv(t)}{dt}$

$\mathcal{L}$ $\underbrace{I(s)}_{Y(s)} = C\,\mathcal{L}\left(\frac{dv(t)}{dt}\right)$

$\updownarrow$?

$V(s)$

$I(s) = C\left(sV(s) - v(0^+)\right)$

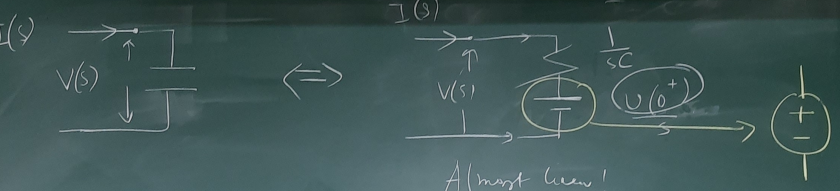
$= sC\,V(s) - C\,v(0^+)$

$V(s) = \frac{1}{sC}I(s) + \frac{v(0^+)}{s}$

$V(t) \leftarrow X(s)$

$f(t)$	$F(s)$
$\frac{df(t)}{dt}$	$sF(s) - f(0^+)$
$\int_0^t f(t)\,dt$	$\frac{F(s)}{s}$

PS7



Almost there!

Implication: Circuit w/ res/cap/batt

(=> res/batt in s-domain

=> solve KCL/KVL in s-domain

PS8

Inductor

$$\Psi(t) = -L\,I(t)$$

$$V(t) = -\frac{d\Psi(t)}{dt}$$

Lenz's law

$$V(t) = L\,\frac{dI(t)}{dt}$$

L: inductance

